

# MLFF Training-Data System Contract Specification

## MLFF-DATA0

mdstats project

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## Scope

This document is the cross-cutting current system contract for the mdstats MLFF training-data and fine-tuning workflow. It owns invariants that span multiple narrower module specifications: evidence-role separation, identity and lineage boundaries, leakage prevention, protocol identity, target/replay separation, MACE realization, calibration applicability, and append-only active-learning lineage.

The legacy filename is retained for stable repository references. It is **not** an implementation stage plan. Developer sequencing and future gates belong in `workplans/`; completed implementation chronology belongs in `docs/history/mlff/`. Narrower current specifications under this directory own module-local API, schema, algorithm, persistence, and runtime details.

## Normative principles

1. Source facts, eligibility, partition, selection, weighting, exposure, and acquisition are separate immutable record families.
2. A frame that has contributed a gradient is not independent validation evidence for that model.

3. Cross-validation SHALL train one fresh model per held-out evaluation fold.
4. A held-out evaluation fold SHALL NOT control stopping or checkpoint choice.
5. Every fold SHALL have a checkpoint-monitor domain disjoint from both its gradient-training domain and held-out evaluation fold.
6. Cross-validation SHALL be bound to the complete `TrainingProtocolIdentity` used for final training. Naive and replay protocols are different protocols.
7. Feature fitting, label-derived difficulty features, E0 fitting, and selection SHALL inspect only the applicable training domain.
8. Locked tests SHALL NOT affect fitting, selection, protocol choice, stopping, checkpointing, uncertainty calibration, or acquisition.
9. Locked tests SHALL be absent from training configurations and activated only after a `ProtocolFreezeRecord` and selected committee identity exist.
10. Partition-critical system-profile features SHALL exist before the outer partition is locked.
11. The first MACE adapter SHALL support one target label domain, an optional replay head, and fixed-file native training only.
12. The first MACE adapter SHALL verify native head ordering, checkpoint control, and target/replay exposure realization against its version lock.
13. Replay retention SHALL be governed by an explicit constraint and applied to saved candidate checkpoints.
14. Dynamic per-epoch resampling SHALL require a custom runtime adapter or explicit multi-job protocol; fixed files alone SHALL NOT claim that feature.
15. Active-learning calibration SHALL be bound to the actual final committee and a declared applicability domain.
16. Existing frame roles SHALL be inherited unchanged by active-learning child datasets unless a new evaluation lineage is explicitly created.

## **Record-family contract**

| Record                         | Owns                                                                         | Must not own                                                          |
|--------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| TrainingDataSource             | source path, hashes, composition, controls, ensemble, quality, label domain  | frame eligibility or split role                                       |
| TrainingFrameRecord            | source-bound frame facts and identity references                             | eligibility, partition, selection, exposure, acquisition              |
| FrameEligibilityDecision       | post-DFT label/quality outcome                                               | partition or training membership                                      |
| PartitionAssignment            | one statistical role under one partition policy                              | feature-selection outcome                                             |
| PartitionFeasibilityReport     | support for requested roles and declared reductions                          | fabricated independent cohorts                                        |
| PartitionIndependenceReport    | evidence grade, purge, correlation, and duplicate limits                     | claim of stronger independence than observed                          |
| SelectionAssignment            | selected/not-selected and reasons                                            | epoch use count                                                       |
| ExposureAssignment             | head, epochs, weights, and use counts                                        | independent-evidence claim                                            |
| MaceExposureRealizationRecord  | actual loader counts, duplication, batches, and property exposures           | intended exposure only                                                |
| AtomicReferenceFitRecord       | training-domain E0 corrections and residual                                  | held-out labels                                                       |
| TrainingProtocolIdentity       | complete model/data/replay/objective/exposure/checkpoint protocol            | mutable runtime observations                                          |
| ProtocolFreezeRecord           | frozen selected protocol and committee identities                            | evaluation results                                                    |
| CandidateAdmissibilityDecision | pre-DFT geometry and trajectory safety                                       | DFT convergence claim                                                 |
| AcquisitionDecision            | calibrated or rank-only acquisition result                                   | post-DFT eligibility                                                  |
| MaterialProfileIdentity        | user-declared phases, geometry, chemistry modifiers, and optional extensions | structural feature values or automatic classification claims          |
| AtomGroupCatalog               | immutable group selectors and phase linkage                                  | per-frame group membership arrays unless owned by a provider artifact |
| ConditionAxisCatalog           | axes and their coverage, material design rules, coverage, and partitioning   | physical features, or partition assignment                            |
| MaterialProfileCatalog         | axes and their coverage, material design rules, coverage, and partitioning   | physical features, or partition assignment                            |

Every serialized record SHALL include a versioned schema and deterministic content digest. Policy and decision records SHALL additionally carry their policy identity and explicit failure reasons where applicable. Identity records SHALL carry their declared parent or provider lineage rather than a fictitious policy digest.

## Identity contract

### Source occurrence

`frame_uid` SHALL be derived from source identity and source frame index. It identifies an occurrence, not a unique physical configuration.

### Geometry and label identities

The following identities SHALL be separate:

```
geometry_fingerprint
label_payload_digest
labeled_configuration_fingerprint
```

`geometry_fingerprint` SHALL exclude energy, force, and stress labels. `label_payload_digest` SHALL include selected labels and label-domain identity. The combined fingerprint SHALL identify the same geometry with the same labeled payload.

Leakage audits SHALL use exact UID overlap, exact geometry overlap, exact labeled-configuration overlap, near-duplicate geometry/descriptor distance, and forbidden temporal proximity.

## Label-domain and energy contract

A label-domain fingerprint SHALL be decomposed into:

```
TheoryIdentity
EnergyReferenceIdentity
DerivativeConvention
NumericalQualityProfile
SoftwareProvenance
```

A versioned compatibility policy SHALL classify differences. Exact equality of all provenance fields is not required; theory- or energy-reference differences cannot be silently combined.

The first adapter SHALL export one target label domain per MACE bundle. Multiple incompatible target domains SHALL produce multiple bundles.

The selected VASP energy channel SHALL be named, complete, and consistent with the derivative labels. The channel identity SHALL be preserved in provenance.

## Atomic-reference contract

The structural `AtomicReferenceIdentifiabilityReport` SHALL record:

```
count matrix
rank
singular values
condition number
null-space dimension
```

identifiable combinations  
policy outcome  
transfer limitations

It SHALL NOT contain fitted corrections or fit residuals.

Each fold and final training domain SHALL receive a separate AtomicReferenceFitRecord containing:

training-domain frame UIDs  
element support  
identifiability-report digest  
foundation-checkpoint digest  
fitted corrections  
fit residual  
solver/tolerance  
policy outcome

The fit SHALL exclude held-out evaluation, checkpoint-monitor, outer monitor, calibration, and locked-test labels. Missing elemental support SHALL fail or invoke an explicit alternative policy.

An emitted MACE configuration SHALL contain the exact version-supported E0 value, normally an explicit atomic-number mapping. A conceptual record name SHALL NOT be written into the E0s field.

## Cell, strain, stress, and virial contract

ASE cell vectors SHALL be treated as rows. Fractional row vectors SHALL map as

$$\mathbf{r}_{\text{row}} = \mathbf{s}_{\text{row}} \mathbf{H}.$$

For reference cell  $\mathbf{H}_0$  and current cell  $\mathbf{H}_t$ , the reported deformation gradient acting on Cartesian column vectors SHALL be

$$\mathbf{F} = (\mathbf{H}_0^{-1} \mathbf{H}_t)^T.$$

The implementation SHALL record whether internal calculations use an equivalent right-acting row-vector map. Tests SHALL include nonsymmetric shear and rotated stretch fixtures.

Canonical REF\_stress SHALL be a symmetric Cartesian 3 x 3 Cauchy-stress tensor in eV/Angstrom<sup>3</sup>, using the ASE/MACE sign convention verified by the version- locked adapter. Virial and stress SHALL use distinct keys.

The export gate SHALL test units, sign, Voigt order, shear factors, and MACE read-back. Missing stress MAY use config\_stress\_weight=0.0 under an explicit heterogeneous-label policy.

## Event, feature, and blinding contract

Full-resolution event detection SHALL run before ordinary thinning. Protected events include coordination changes, site changes, ring crossings, topology changes, strain extrema, and high but physical restoring-force excursions.

DATA4 SHALL provide partition-critical system-profile features before the outer partition is locked. For LTA these SHALL include resolvable coarse ring-site, off-center, coordination-change, site-change, ring-crossing, and framework- integrity states.

Raw feature providers SHALL be partition-independent. Scaling, PCA, whitening, or fitted metrics SHALL be represented by separate static templates and fitted records:

FeatureMetricPolicyTemplate  
FoldFeatureTransform  
FoldFeatureMetricFit  
FinalFeatureTransform  
FinalFeatureMetricFit

A static template SHALL define feature blocks, scaling rules, block/species weights, retained dimensions, missing-block behavior, distance metric, dtype, and tolerance. A fitted record SHALL contain only parameters learned from its declared training domain.

Foundation descriptors may be computed for all domains. Foundation-model residuals require DFT labels and SHALL be exposed only in a TrainingDifficultyFeatureCatalog for the applicable training domain. Outer monitor, calibration, held-out-fold, and locked-test residuals SHALL remain blinded until authorized evaluation.

## Partition feasibility and outer-role contract

A PartitionRoleBudgetPolicy SHALL declare requested roles, minimum block counts, minimum independence grades, and allowable reductions. A PartitionFeasibilityReport SHALL classify support before assignment.

Allowed outcomes include:

fully\_supported  
supported\_with\_temporal\_blocks\_only  
calibration\_deferred  
challenge\_set\_external\_only  
reduced\_cross\_validation\_folds  
insufficient\_for\_locked\_test  
insufficient\_for\_requested\_roles

The workflow SHALL NOT fabricate every role from a short trajectory to satisfy fixed percentages.

Every feasible target label domain may define:

development\_pool  
outer\_monitor\_validation  
uncertainty\_calibration  
locked\_interpolation\_test  
zero or more locked\_challenge\_tests

The calibration domain MAY be deferred to later independent calculations. Locked tests SHALL remain operationally sealed.

The LTA profile SHALL use hierarchical applicable schemas rather than a global Cartesian product:

unstrained: composition x temperature x regime  
strained: composition x reference-condition x strain-mode x sign x regime

Every cohort SHALL emit a PartitionIndependenceReport with machine-readable grades such as independent replica, independent ordering, independent run, purged temporal block, slow-state not decorrelated, or insufficient independence.

## Cross-validation and training-protocol contract

A TrainingProtocolIdentity SHALL bind:

foundation checkpoint and head  
naive or multi-head mode  
replay source, selection, and monitor  
training objective and property weights  
target/replay head weights  
exposure backend and balancing policy  
checkpoint metric and checkpoint-control policy  
replay-retention policy  
optimizer, scheduler, epoch cap, and seed policy  
MACE adapter lock

A CrossValidationJobFamily SHALL contain  $K$  independent jobs using the same complete protocol.  
For fold  $k$ :

1. designate a held-out evaluation fold;
2. form the non-evaluation domain;
3. carve a deterministic purged checkpoint-monitor split;
4. fit transform, metric, E0, difficulty features, and selector only on the remaining gradient-training domain;
5. initialize a fresh model and optimizer;
6. train under the declared MACE checkpoint-control policy;
7. select and freeze a checkpoint without inspecting the held-out fold;
8. evaluate only then on the held-out fold;
9. emit out-of-fold predictions and independence grades.

A rotating inner-validation fold inside one evolving model is prohibited. Cross-validation results from a naive protocol SHALL NOT be used as validation of a replay protocol.

## Selection-budget contract

The selector SHALL construct one deterministic master order with mandatory anchors followed by quota-interleaved evidence classes:

representative coverage  
species-environment coverage  
rare events  
descriptor FPS  
difficulty enrichment

SelectionBudgetPolicy SHALL define counts or fractions and a deterministic deficit-redistribution rule. Later evidence classes SHALL NOT be starved because an earlier selector consumed the size budget.

Near-duplicate pruning SHALL occur during master-order construction. Requested sizes SHALL be exact prefixes. A size smaller than the mandatory-anchor count is infeasible.

Species-specific Li, Na, and K environment selection SHALL be available in the LTA profile. Whole-cell framework-dominated descriptors alone are insufficient.

## Training objective, weights, and checkpoint metrics

TrainingObjectivePolicy SHALL define the loss family, energy/force/stress weights, head weights, normalization, robust-loss settings, and missing-label behavior.

ConfigurationWeightPolicy and PropertyWeightPolicy SHALL define condition-, regime-, event-, quality-, and property-specific weights. Selection, weighting, and exposure are separate decisions.

The first adapter SHALL use standard MACE configuration/property weights and SHALL NOT claim a species-aware atomwise loss. It SHALL report species-resolved force metrics and impose profile focus-group acceptance or checkpoint constraints. Any custom species-aware loss SHALL create a different TrainingProtocolIdentity.

CheckpointMetricPolicy SHALL define the primary target scalar plus energy, force, stress, species, worst-condition, rare-event, and replay-retention constraints.

## Exposure and MACE realization contract

Exposure backends are:

NATIVE\_MACE\_FIXED  
CUSTOM\_EPOCH\_RESAMPLE  
MULTI\_JOB\_RESAMPLE  
FINAL\_REFIT

The first adapter SHALL support only NATIVE\_MACE\_FIXED. Dynamic epoch resampling SHALL NOT be represented by static files alone.

Every run SHALL emit a MaceExposureRealizationRecord containing:

real\_pt\_data\_ratio\_threshold  
pre-MACE target/replay counts  
post-MACE effective target/replay counts  
implicit duplication factor  
expected and observed batches  
configuration, energy, force, and stress exposures

The adapter SHALL disable implicit target duplication where supported. Otherwise it SHALL bind the realized duplication to TrainingProtocolIdentity and fail if observed loader counts differ from the accepted plan.

## MACE checkpoint-control and replay contract

The initial adapter target is mace-torch==0.3.16, with a tested compatibility matrix. It SHALL capture package digest, source/tag identity, CLI help, and key parser, loader, and training-loop source digests.

MACE 0.3.16 uses the last validation head for native scheduling, patience, and best-checkpoint decisions. The accepted first mode is:

NATIVE\_TARGET\_LAST\_WITH\_EXTERNAL\_CONSTRAINT\_AUDIT

The adapter SHALL:

1. verify that the target monitor is the last validation head;
2. prevent replay-head behavior from terminating training;

3. retain every evaluation checkpoint;
4. externally evaluate target, focus-group-resolved, and replay monitors;
5. apply CheckpointMetricPolicy deterministically;
6. fail closed when version-tested behavior changes.

Replay preparation SHALL precede replay-aware cross-validation. The replay monitor SHALL be disjoint from replay training. ReplayRetentionPolicy SHALL define baseline, metric, tolerated degradation, and failure/override behavior.

## **MACE artifact contract**

Extended XYZ SHALL contain only MACE-readable labels, weights, and compact stable identities. Complete provenance and selection reasons SHALL live in a sidecar manifest keyed by `frame_uid`.

The development bundle SHALL contain no locked-test path and SHALL include the complete TrainingProtocolIdentity, objective, checkpoint metric, replay plan, checkpoint-control policy, E0 mapping, and adapter lock.

The following artifacts SHALL be separate:

```
development_bundle/  
calibration_bundle/  
sealed_evaluation_bundle/  
evaluation_activation/  
evaluation_results/
```

A sealed evaluation bundle MAY be prepared early. An EvaluationActivationDecision SHALL require a ProtocolFreezeRecord, selected committee identity, complete training-protocol digest, and checkpoint-selection decision.

## **Calibration and active-learning contract**

Out-of-fold predictions MAY diagnose uncertainty ranking. Numerical thresholds SHALL be calibrated from the actual final committee on a dedicated calibration cohort.

CalibrationApplicabilityDomain SHALL record covered elements, compositions, temperatures, strains, cell sizes, site/event classes, descriptor-distance range, force/stress range, and framework-integrity state.

CalibrationTransferDecision SHALL classify candidates as:

```
within_calibrated_domain  
rank_only_outside_domain  
recalibration_required  
rejected_incompatible_domain
```

Locked tests are forbidden from calibration and acquisition.

Pre-DFT candidates receive CandidateAdmissibilityDecision. Child datasets SHALL inherit existing roles unchanged by default. Selection-biased new labels enter the development pool; independent random labels may form new validation/calibration cohorts; predeclared challenges may form new locked challenge sets. A full repartition SHALL create a new evaluation lineage.

## **Authority and supersession**

This specification owns only the cross-cutting invariants stated above. The dedicated current specifications listed in `README.md` own their module-local behavior and may refine implementation details without weakening these invariants. Runtime/product gates remain normative where they define current software behavior; developer implementation gates are non-normative coordination artifacts under workplans/.

The complete pre-DOC-GOV1 mixed specification, including completed stage chronology and historical/future developer planning, is preserved at `docs/history/mlff/manual_snapshots/mlff_data_stage_plan_spec_pre_doc_gov1.md` and is non-normative.